

Daily Content Intel

August 10, 2026

Topic A — Reel

TELEPROMPTER COPY (paste into Instagram)

Okay, if your athlete just started football or lacrosse preseason, this one matters this week.

New prospective cohort study. 299 elite footballers, GPS units on them across two full seasons, 67 hamstring strains. What separated the injured players from the healthy ones was the size of the week to week jump. Weekly high speed running distance, weekly change in total distance, weekly change in high speed running, all three showed up as significant.

Muscle architecture came along with it. Shorter biceps femoris fascicles and a steeper pennation angle went with

higher risk, and longer fascicles went with lower risk.

Here's the honest part. Strength and architecture together explained 12% of the risk, adding running load got it to 20%, which leaves about 80% unexplained. No screen predicts a hamstring strain. Anybody telling you theirs does is guessing.

What I'd do Monday. Build sprint exposure up in steps across preseason instead of jumping straight to full speed on day one, and keep loading the hamstring long and heavy, Nordics and Romanian deadlifts through a big range, all season and not just in July.

Be Good. Help Someone. Learn Lots.

Hook: 299 pro footballers, 2 full seasons, 67 hamstring strains, and the thing that got them was the week-to-week jump.

Spoken script (word for word):

Okay, if your athlete just started football or lacrosse preseason, this one matters this week.

New prospective cohort study. 299 elite footballers, GPS units on them across two full seasons, 67 hamstring strains. What separated the injured players from the healthy ones was the size of the week to week jump. Weekly high speed running distance, weekly change in total distance, weekly change in high speed running, all three showed up as significant.

Muscle architecture came along with it. Shorter biceps femoris fascicles and a steeper pennation angle went with higher risk, and longer fascicles went with lower risk.

Here's the honest part. Strength and architecture together explained 12% of the risk, adding running load got it to 20%, which leaves about 80% unexplained. No screen predicts a hamstring strain. Anybody telling you theirs does is guessing.

What I'd do Monday. Build sprint exposure up in steps across preseason instead of

jumping straight to full speed on day one, and keep loading the hamstring long and heavy, Nordics and Romanian deadlifts through a big range, all season and not just in July.

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On-screen text seeds:

- 0:00 299 players. 2 seasons. 67 hamstring strains.
- 0:08 The size of the week-to-week jump
- 0:20 Weekly change in high-speed running = higher risk
- 0:34 Short fascicles, steep pennation angle = higher risk
- 0:48 12% explained by strength + architecture
- 0:54 20% once you add running load
- 1:00 80% still unexplained
- 1:12 Monday: ramp sprint exposure in steps
- 1:20 Monday: load the hamstring long, all season

Caption (copy-paste ready):

Preseason is where hamstrings get written off for the year.

299 elite footballers, tracked with GPS across two full seasons, 67 hamstring strains. The players who got hurt had bigger jumps from one week to the next. Weekly change in total distance and weekly change in high-speed running both separated the injured group from the healthy one.

Muscle architecture showed up too. Shorter biceps femoris fascicles and steeper pennation angles went with higher risk.

The part I appreciate about this paper: strength plus architecture explained 12% of the risk, adding load exposure got it to 20%, and the authors say plainly that about 80% is still unexplained. That is what honest risk research looks like.

So no screen predicts a hamstring strain. What you can control is the ramp and the strength.

Build sprint exposure in steps through camp. Keep training the hamstring long and heavy into November, not just August.

Be Good. Help Someone. Learn Lots.

Dr. Lars Stevenson, PT, DPT

threshold.clientsecure.me

#hamstring #footballtraining #lacrosse

#speedtraining #athletedevelopment

#strengthandconditioning #sportsPT

#injuryprevention #highschoolfootball

#returntosport #nova #restonva

Personalize this

- Which of your football or lacrosse athletes came into camp this month with the biggest gap between their July running and their week-1 practice load, and what did you change in their week to keep the jump small?
- You've made return-to-sport calls on hamstring strains before. What does your criteria set look like when the research says 80% of the risk is unexplained, and how do you explain that honestly to a parent who wants a date?
- Where does the hamstring sit in your CROSS framework for a preseason field-sport athlete, and what's the actual

Nordic and RDL progression you're running right now at Threshold?

Screenshots:

- title | <https://pmc.ncbi.nlm.nih.gov/articles/PMC12644382/> | "The Relationship Between Running Load, Strength, Muscle Architecture and Hamstring Strain Injury Across Two Seasons of Elite Male Australian Football: A Prospective Cohort Study"
- physiology | <https://pmc.ncbi.nlm.nih.gov/articles/PMC12644382/> | "Combining strength and architectural variables of BF_{th} pennation angle, fascicle length and peak force was able to explain 12% of variance in the risk of sustaining a HSI."
- actionable | <https://pmc.ncbi.nlm.nih.gov/articles/PMC12644382/> | "Practitioners should appropriately plan and periodise player running loads to avoid large changes in distance"

Sources:

- <https://pmc.ncbi.nlm.nih.gov/articles/PMC12644382/>

- <https://link.springer.com/article/10.1186/s40798-025-00944-4>

Topic B — Reel

TELEPROMPTER COPY (paste into Instagram)

Okay, hot take. 220 minus your age needs to leave the weight room.

New study, 4,375 endurance athletes, their real max heart rate from actual training and racing compared against the age formulas. The formulas came in low by about 5 to 6 beats on average, and the limits of agreement ran from 20 beats below to 9 beats above. That's a 29 beat window on the one number every zone underneath it is built from.

With young athletes it runs the other way. In 147 team sport kids across soccer, futsal, basketball and water polo, 220 minus age came in higher than their measured max. So you're chasing a 15 year old toward a ceiling they don't actually have, and calling it conditioning.

The authors' words: age based formulas should be abandoned as primary tools.

So measure it. Warm up 5 to 8 minutes, run a hard progressive test in the mode they actually train in, chest strap on, and take the highest number you see. That's their max until something better shows up.

Does that make sense?

Be Good. Help Someone. Learn Lots.

Hook: If you're setting your athlete's conditioning off 220 minus their age, you're running the season on a number somebody eyeballed in 1970.

Spoken script (word for word):

Okay, hot take. 220 minus your age needs to leave the weight room.

New study, 4,375 endurance athletes, their real max heart rate from actual training and racing compared against the age formulas.

The formulas came in low by about 5 to 6 beats on average, and the limits of agreement ran from 20 beats below to 9 beats above. That's a 29 beat window on the one number every zone underneath it is built from.

With young athletes it runs the other way. In 147 team sport kids across soccer, futsal, basketball and water polo, 220 minus age came in higher than their measured max. So you're chasing a 15 year old toward a ceiling they don't actually have, and calling it conditioning.

The authors' words: age based formulas should be abandoned as primary tools.

So measure it. Warm up 5 to 8 minutes, run a hard progressive test in the mode they actually train in, chest strap on, and take the highest number you see. That's their max until something better shows up.

Does that make sense?

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On-screen text seeds:

- 0:00 220 minus your age
- 0:06 A number from 1970
- 0:16 4,375 athletes tested against the formula
- 0:26 Off by 20 beats low to 9 beats high
- 0:34 A 29-beat window
- 0:44 In youth athletes it runs HIGH
- 0:56 "Age-based formulas should be abandoned"
- 1:06 Warm up. Progressive max test. Chest strap.
- 1:16 Take the highest number you see.

Caption (copy-paste ready):

220 minus your age is a placeholder, and most programs treat it like physiology.

A new analysis of 4,375 endurance athletes compared real max heart rates from training and racing against the age formulas. On average the formulas came in 5 to 6 beats low, and the limits of agreement ran roughly 20 beats below to 9 beats above. That is a 29-beat spread on the single number your zones are built from.

In young team-sport athletes the error flips. Across 147 kids in soccer, futsal, basketball and water polo, 220 minus age read higher than their measured max, so the target you're handing a 15-year-old is a ceiling they don't have.

The authors put it plainly: age-based formulas should be abandoned as primary tools.

Measure it instead. 5 to 8 minute warm-up, a hard progressive effort in the mode they actually train in, chest strap rather than a wrist optical sensor, and take the highest number you see. Anyone with cardiovascular symptoms or medical risk gets a supervised test or skips this.

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Dr. Lars Stevenson, PT, DPT
threshold.clientsecure.me

#conditioning #heartratetraining
#athletedevelopment
#strengthandconditioning #footballtraining
#lacrosse #youthsports #sportsscience
#coachingeducation #highschoolathlete

#nova #restonva

Personalize this

- What are you actually using to set conditioning intensity for your football and lacrosse athletes right now, and have you ever run a real max test on one of them and seen how far off the formula was?
- You've got athletes returning from injury where conditioning has to be dosed carefully. How would a 29-beat error in their assumed max change what you'd prescribe in weeks 4 through 8 of a return-to-sport block?
- Threshold's whole pitch is measuring instead of assuming. What's one other number in a typical HS program that everybody inherits without testing, and what do you test in its place?

Screenshots:

- title | <https://www.frontiersin.org/journals/sports-and-active-living/articles/10.3389/fspor.2026.1806303/full> | "An exploratory study of maximal heart rate determination in endurance athletes: laboratory testing

vs. field based"

- physiology | <https://www.frontiersin.org/journals/sports-and-active-living/articles/10.3389/fspor.2026.1806303/full> | "age-based HRmax formulas show systematic underestimation and wide individual error in endurance athletes"
- actionable | <https://www.frontiersin.org/journals/sports-and-active-living/articles/10.3389/fspor.2026.1806303/full> | "These findings support the use of individualized, context-specific HRmax assessment, while highlighting the limitations of relying solely on age-based predictions."

Sources:

- <https://www.frontiersin.org/journals/sports-and-active-living/articles/10.3389/fspor.2026.1806303/full>
- <https://pubmed.ncbi.nlm.nih.gov/25114367/>

Reference

Runner-up topics

- Resisted sprint training meta-analysis (Frontiers in Physiology): sled loads improve 5, 10 and 20 m sprint plus change of direction. Straight speed-development Reel for football and lacrosse.
- Iron Radio ep. 247 on the FDA advisory panel backing six compounding peptides including BPC-157: a hot take on why HS athletes should stay away from the grey market, framed around what actually builds recovery.
- Mike Nelson's creatine myths PDF (kidney markers, the 2009 hair-loss rugby study, the caffeine interaction study from 1996): a myth-busting Reel for parents of youth athletes, needs a fresh primary source since the registry already carries creatine items from 7/2 and 7/28.

Sources scanned today

- Newsletter digest
newsletter_digest_2026-08-10.pdf (3

sources): Mike T. Nelson Weekly Wrap Up (creatine myths PDF, "220 minus age" max HR breakdown citing Fox & Haskell 1970, Robergs & Landwehr 2002, Tanaka 2001, Lach 2021), Iron Radio ep. 247 (FDA peptide advisory panel, BPC-157 evidence), 2 Dr. Mercola sends (pure promo, skipped).

- PubMed / PMC: recent sprint, plyometric, hamstring, and ACL work in adolescent and team-sport athletes. Best new athlete-facing item was the Breed et al. two-season AFL prospective cohort on running load + hamstring architecture (Sports Medicine - Open, 2025).
- Frontiers in Sports and Active Living: Ausland, Kelemen & Seiler max heart rate lab vs field study (April 2026, N = 4,375).
- Frontiers in Physiology: resisted sprint training meta-analysis (held as runner-up).
- BJSM / JOSPT / Sports Health: nothing newer than what the registry already carries this week.

- ScienceDirect youth-athlete sleep paper (S2211266926000058): blocked, HTTP 403. Skipped.
- Reddit r/AdvancedFitness: nothing surfacing a new primary source this week.